0

Below is an exerpt of the ladder logic to control the pick and place device shown in the simulation. Logic related to other parts of the machine have been removed. The 'main' part of this code is from line 1 to line 532 (ED). Below line 532 (ED) are 11 subroutines.

The pick and place device consist of 3 servor motors, 2 pnuematic solenoids and 1 laser sensor (analog input).

The PLC used in this project is a Panasonic FP-XH C60TD with an analog input cassette (AFPX-AD2). The antique software used to write the ladder logic is called FPWIN GR, which will be discontinued on September 30, 2024.

The first line Y19 is the contactor that controls the main power for the pick and place device. R1 never turns off.

R1 ─┤/├─ Y19 [ ] TB1

2

Pneumatic solenoids: Y14 and Y10
Y10 controls the end effector
Y14 ejects misplaced belt loops from the device

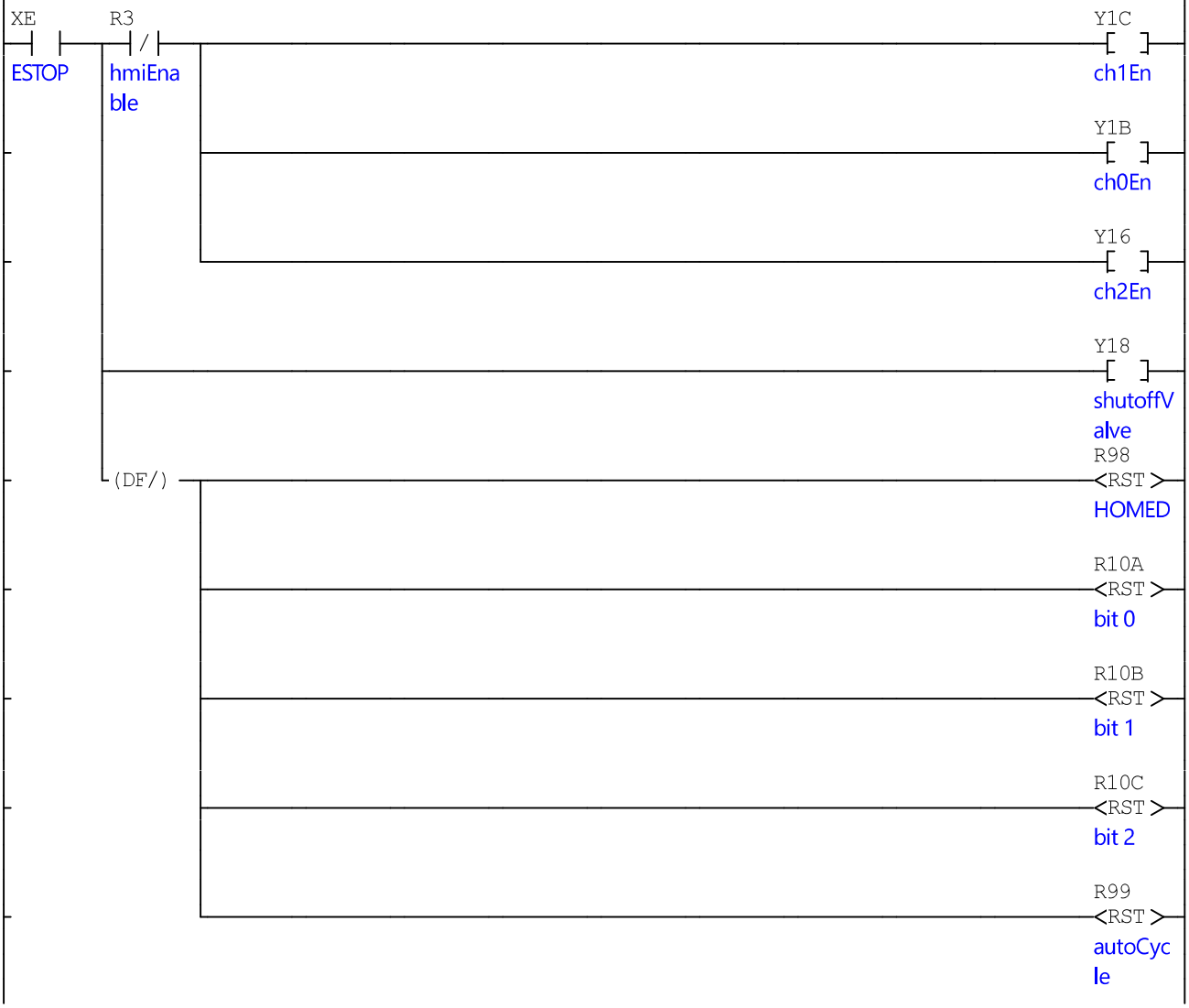
R97 ─┤ ├─ Y14 [ ] AirJetOut

4

R155 ─┤ ├─ Y10 [ ] sol1picker
hmi pinch
R11B ─┤ ├─
autoPick

7

Emergency stop: XE
When the emergency stop is pressed -
-all the servos are disabled (Y1C, Y1B, Y16)
-the main air is shut off (Y18)
-the device must be rehomed (R98)
-the pick and place cycle is reset and turned off (R10A, R10B, R10C, R99)
{in FPWIN GR: DF = leading edge of a signal and DF/ = trailing edge of a signal}



32 Pick and place cycle shown in the simulation.

The pick and place cycle consist of 8 states. The cycle starts at state 1 and increments +1 to state 8 where the cycle ends and starts again at state 1.
[state 1 -> state 2 -> ... -> state 8 -> state 1 -> state 2 -> ...]

Each state is described using 3 relays (R10A, R10B, R10C).
For example:
R10A = off, R10B = off, R10C = off
State 000: State 1
or
R10A = off, R10B = off, R10C = on
State 001: State 2
If all relays are on (on, on, on = State 111 = State 8)

The states follow this order: 000, 001, 010, 011, 100, 101, 110, 111
A brief description of each state:
000: Measure with laser the height of fabric, index magazine if it is empty
001: Index X servo right
010: Index Y servo down
011: Turn end effector on
100: Index Y servo up
101: Index X servo left
110: Turn end effector off
111: Restart cycle

From the HMI the cycle is started, there are 2 options on the HMI:
automatic mode (R99)
step mode (R127)[when R127 is on, another button is available on the HMI to step through each state manually (R126)]

R98 (HOMED) — R9F (hmiON) — (DF) — R99 (autoCycle) <SET>

38 C1022 (conEmptyCount) — (DF) — R99 (autoCycle) <RST>
R128 (hmiOFF) — (DF/)

47 R98 (HOMED) — R126 (stepTrigHmi) — (DF) — R129 (stepTrig) [ ]

51 State 1: 000: Measure with laser the height of fabric, index magazine if empty
R94 sets off a series of events that measures the stack of fabric.
-If the laser measurement is empty, the magazine is indexed 1 position, and R94 is triggered again. This continues until a non empty measurement is taken or until every magazine cartridge has been checked. If all cartridges measure empty, the cycle shuts off and must be restarted by the operator.
-If the laser measurement is not empty, bit 2(R10C) is set and the program moves to State 2.

R99 (autoCycle) — R10A (bit 0) — R10B (bit 1) — R10C (bit 2) — XB (ch1 safe) — R10F (wait) — (DF) — R94 (measureTrig) [ ]
R127 (stepScreen) — R129 (stepTrig) — R12A (measuring) <SET>

```
65  R10F      R911E
    wait      ch2 sig      (DF/)                                          R112
                                                                          <SET>
                                                                          waitTim
                                                                          er
72  R112                                                   TMR     1 , K   10
    waitTim
    er
76  T1                                                                    R10F
                                                                          <RST>
                                                                          wait
                                                                          R112
                                                                          <RST>
                                                                          waitTim
                                                                          er
83  R99       R10A      R10B      R10C      R103      R12A      TML     13 , K   5      R10C
    autoCyc   bit 0     bit 1     bit 2     notEmp    measuri   measur                 <SET>
    le                                      ty        ng        e                      bit 2
    R127                                                                               R12A
    stepScr                                                                            <RST>
    een                                                                                measuri
                                                                                       ng
99  State 2: 001: Index X servo right
    R10D triggers a subroutine to move the X servo right to a predetermined distance
    When the X servo stops moving (R911C), the program moves to State 3.

    R99                 R10A      R10B      R10C      XB                             R10D
    autoCyc             bit 0     bit 1     bit 2     ch1 safe  (DF)                 [ ]
    le                                                                               001right
    R127      R129
    stepScr   stepTrig
    een
109 R119      R911C
    wait      ch0 sig      (DF/)                                          R119
                                                                          <RST>
                                                                          wait
                                                                          R10B
                                                                          <SET>
                                                                          bit 1
                                                                          R10C
                                                                          <RST>
                                                                          bit 2
122 State 3: 010: Index Y servo down
    R10E triggers a subroutine to move the Y servo down based off the laser
    measurement. When the Y servo stops moving (R911D), the program moves to State 4

    R99                 R10A      R10B      R10C      XB                             R10E
    autoCyc             bit 0     bit 1     bit 2     ch1 safe                       [ ]
    le                                                                               010dow
    R127      R129                                                                   n
    stepScr   stepTrig
    een
```

131 R11A (wait) R911D (ch1 sig) (DF/) → R11A <RST> (wait); R10C <SET> (bit 2)

141 State 4: 011: Turn end effector on
The end effector(R11B) contains 30 needles. When actuated, the needles eject out of the end effector, stabbing the fabric. The amount the needles eject out of the end effector is adjustable. If the needles stick out too far, multiple belt loops are picked up. If the needles stick out not enough, the belt loop will not be picked up/sometimes be picked up. This adjustment must be done physically at the machine. An improvement to this machine would be to have the end effector adjustable electronically - this does not exist - that I'm aware of. When the 0.2s timer (TMR 3) goes off, the program moves to State 5.

XE (ESTOP) R127 (stepScreen) R10A (bit 0) R10B (bit 1) R10C (bit 2) (DF) → R11B <SET> (autoPick); R11C <SET> (pickTimer)
R127 (stepScreen) R129 (stepTrig)

156 R11C (pickTimer) TMR 3 , K 20 → R11C <RST> (pickTimer); R10A <SET> (bit 0); R10B <RST> (bit 1); R10C <RST> (bit 2)

172 State 5: 100: Index Y servo up
R11D triggers a subroutine to move the Y servo up based off the laser measurement. When the Y servo stops moving (R911D), the program moves to State 6

XE (ESTOP) R127 (stepScreen) R10A (bit 0) R10B (bit 1) R10C (bit 2) → R11D [ ] (100up)
R127 (stepScreen) R129 (stepTrig)

181 R11E (wait) R911D (ch1 sig) (DF/) → R11E <RST> (wait); R10C <SET> (bit 2)

191

State 6: 101: Index X servo left
R11F triggers a subroutine to move the X servo left to a predetermined distance
Before the X servo stops moving (R911C), the program moves to State 7.
[The device attempts to release the belt loop while in motion]
{DT90348 is the data register that stores the X servo motor position. In the final position DT90348 = 50000, the inequality sets the next state when DT90348 > 48730, just before the X motor reaches its final position}

R99 (autoCycle) — R10A (bit 0) — R10B (bit 1) — R10C (bit 2) — XB (ch1 safe) — X13 (clear?(op)) — X14 (clear?(mach)) — R11F (101Left)

R127 (stepScreen) — R129 (stepTrig)

202

R120 (wait) — R911C (ch0 sig) — D > DT 90348 , K 48730 (ch0 count)

- R120 <RST> wait
- R10B <SET> bit 1
- R10C <RST> bit 2

223

State 7: 110: Turn end effector off
When the end effector is turned off (R11B), the needles retract which releases the fabric. The program moves to State 8.

R99 (autoCycle) — R10A (bit 0) — R10B (bit 1) — R10C (bit 2) — XB (ch1 safe)

- R11B <RST> autoPick
- R10C <SET> bit 2

R127 (stepScreen) — R129 (stepTrig)

237 State 8: 111: Restart cycle
In this state the program checks to see if the X and Y motors are in the correct position (DT90348 and DT90352) to restart the cycle. If the motors are not in the correct position, wait. If they are in the correct position, the program resets to Step 1.

R99 (autoCycle) | R10A (bit 0) | R10B (bit 1) | R10C (bit 2) | XB (ch1 safe) | D = DT 90348 , K 50000 (ch0 count) → 1

R127 (stepScreen) | D = DT 90352 , K 50000 (ch1 count) ← 1

→ R10A (bit 0) <RST>
→ R10B (bit 1) <RST>
→ R10C (bit 2) <RST>

270 go to jog screen

R100 (hmiJogScreen) | XE (ESTOP) → (CALL 0) (jog screen)

274 Home picker: Call 1 - Call 4, below lines to Call 5 are related to homing-------

R0 (hmiHomeScre) | XE (ESTOP) → (CALL 1) (home screen)

R124 (home seq)

279 R510 (hmiCh0home) (DF) → R536 (ch0 home trig) <SET>

R124 (home seq) | R2 (ch1 homing) (DF/) → R400 (ch0homing) <SET>

```
291  R410 ─┤ ├─(DF)──┬──────────────
```

353 Call 5: get measurement from laser and convert measurement to ch 1 steps (vertical servo calculate steps)

R118 (DF) → 1
hmi measure
R94
measureTrig

1 → [F0 MV , WX 10 , DT 2100] → R156 <SET>
laser input
laser value
calcCh1dist

364 R156 (CALL 5)
calcCh1dist

367 This section keeps track of ch 2 position (CT1020 & CT1021) and moves ch2 when a tray is empty (when cycling). CT1022 keeps track of how many consecutive empty signals were given, after 16, stop running.

R98 HOMED, R99 autoCycle, R157 empty, C1020 emptyRight → CT 1020 emptyRight SV 1020
R127 stepScreen
R9B resetCount
R9E homeResetCoun

380 R98 HOMED, R99 autoCycle, R157 empty, C1020 emptyRight → CT 1021 emptyLeft K 16
R127 stepScreen
R9B resetCount
R9E homeResetCoun

```
393  R98 HOMED | R99 autoCycle | R157 empty | R127 stepScreen | CT 1022 conEmptyCount K 30
     R103 notEmpty (DF)
     R9E homeResetCoun
     R9F hmiON (DF)

408  R157 empty | R10F wait <SET>

412  C1020 emptyRight (DF) | R9A ch2rightEmpty <SET>

418  C1021 emptyLeft (DF) | R9A ch2rightEmpty <RST>
     R9B resetCount [ ]

425  R9B resetCount [F0 MV , K 15 , SV 1020 ]

431  R98 HOMED | R99 autoCycle | R127 stepScreen | R9A ch2rightEmpty / | R157 empty (DF) | R9C ch2Rempty [ ]
     R157 empty <RST>

442  R98 HOMED | R99 autoCycle | R127 stepScreen | R157 empty (DF) | R9D ch2Lempty [ ]
     R157 empty <RST>
```

452 Call 6: Ch1 index down (using laser measurement)

R407 (DF) → R90 <SET> indexDown
hmiCh1indD

R10E (DF) R10E → R11A <SET> wait
010down 010down

464 R90 (CALL 6)
indexDown

467 Call 7: Ch 1 index up (using laser measurement)

R406 (DF) → R91 <SET> indexUp
hmiCh1ndU

R11D (DF) R11D → R11E <SET> wait
100up 100up

479 R91 (CALL 7)
indexUp

482 Call 8: Ch0 index right

R506 (DF) → R92 <SET> indexRight
hmiCh0ndR

R10D (DF) R10D → R119 <SET> wait
001right 001right

494 R92 (CALL 8)
indexRight

497 Call 9: Ch0 index left

R507 (DF) → R93 <SET> indexLeft
hmiCh0ndL

R11F R11F → R120 <SET> wait
101Left 101Left

507 R93 (CALL 9)
indexLeft

510 Call 10: Ch2 index right

R107 (DF) hmiCh2R → R95 <SET> ch2indexR

R9C (DF) ch2Rempty

518 R95 ch2indexR → (CALL 10) ch2indexR

521 Call 11: Ch2 index left

R106 (DF) hmiCh2L → R96 <SET> ch2indexL

R9D (DF) ch2Lempty

529 R96 ch2indexL → (CALL 11) ch2indexL

532 (ED)

533 Sub 0: Manually move picker servos from jog screen on HMI
Set motor speed - Jog motor - 6 possible directions

(SUB 0) jog screen

534 R122 (DF) hmiCh1JogD → 1

R121 (DF) hmiCh1JogU

1 → [F1 DMV , K 10000 , DT 4004] Ch1'w's peed → R98 <RST> HOMED

```
549  R502
     ─┤ ├─(DF)──────────────────────────────────────────────────────────→ 1
     hmiCh0
     JogL
     R501
     ─┤ ├─(DF)──┐
     hmiCh0
     JogR
                                                                          R98
     1 ──→─[F1  DMV      ,   K 10000    ,   DT  5004    ]──────────────<RST>
                                          Ch0'w's                         HOMED
                                          peed
564  R102
     ─┤ ├─(DF)──────────────────────────────────────────────────────────→ 1
     hmiCh2
     JogR
     R101
     ─┤ ├─(DF)──┐
     hmiCh2
     JogL
                                                                          R98
     1 ──→─[F1  DMV      ,   K 10000    ,   DT  2004    ]──────────────<RST>
                                          Ch2'w's                         HOMED
                                          peed
579  R121
     ─┤ ├─[F1  DMV      ,   H  122     ,   DT  4000    ]
     hmiCh1                               Ch1 con
     JogU                                  code

     [F1  DMV      ,   DT  4004    ,   DT  4002    ]
                     Ch1'w's         Ch1'r'sp
                     peed            eed

     [F172  PLSH   ,   DT  4000    ,   K  1        ]
                     Ch1 con
                      code
599  R122
     ─┤ ├─[F1  DMV      ,   H  113     ,   DT  4000    ]
     hmiCh1                               Ch1 con
     JogD                                  code

     [F1  DMV      ,   DT  4004    ,   DT  4002    ]
                     Ch1'w's         Ch1'r'sp
                     peed            eed

     [F172  PLSH   ,   DT  4000    ,   K  1        ]
                     Ch1 con
                      code
619  R502
     ─┤ ├─[F1  DMV      ,   H  113     ,   DT  5000    ]
     hmiCh0                               Ch0 con
     JogL                                  code

     [F1  DMV      ,   DT  5004    ,   DT  5002    ]
                     Ch0'w's         Ch0'r'sp
                     peed            eed

     [F172  PLSH   ,   DT  5000    ,   K  0        ]
                     Ch0 con
                      code
```

```
639 R501 ─┤ ├──[F1  DMV      ,  H  122     ,  DT  5000   ]
    hmiCh0                                   Ch0 con
    JogR                                      code
        [F1  DMV      ,  DT  5004    ,  DT  5002   ]
                       Ch0'w's        Ch0'r'sp
                       peed           eed
        [F172  PLSH   ,  DT  5000    ,  K  0       ]
                       Ch0 con
                        code
659 R102 ─┤ ├──[F1  DMV      ,  H  1113    ,  DT  2000   ]
    hmiCh2                                   Ch2 con
    JogR                                      code
        [F1  DMV      ,  DT  2004    ,  DT  2002   ]
                       Ch2'w's        Ch2'r'sp
                       peed           eed
        [F172  PLSH   ,  DT  2000    ,  K  2       ]
                       Ch2 con
                        code
679 R101 ─┤ ├──[F1  DMV      ,  H  1122    ,  DT  2000   ]
    hmiCh2                                   Ch2 con
    JogL                                      code
        [F1  DMV      ,  DT  2004    ,  DT  2002   ]
                       Ch2'w's        Ch2'r'sp
                       peed           eed
        [F172  PLSH   ,  DT  2000    ,  K  2       ]
                       Ch2 con
                        code
699 ──────────────────────────────────────────────────────────────(    RET    )──

700     Sub 1: Home screen
        --Inside Sub 1:
        - Sub 2: Ch0 home
        - Sub 3: Ch1 home
        - Sub 4: Ch2 home

    ──────────────────────────────────────────────────────────────(SUB     1)──

701 R400 ─┤ ├──────────────────────────────────────────────────────(CALL    2)──
    ch0hom                                                         ch0Hom
    ing                                                            eSeq
704 R2 ─┤ ├──────────────────────────────────────────────────────(CALL    3)──
    ch1 ho                                                         ch1Hom
    ming                                                           eSeq
707 R402 ─┤ ├──────────────────────────────────────────────────────(CALL    4)──
    ch2 ho                                                         ch2Hom
    ming                                                           eSeq
```

710 sub 2: homing sequence - ch0
R511: step 1 jog left, hit sensor
R509: step 2 index right
R514: step 3 jog slower left, hit sensor end
R515: step 4 move to offset set by hmi -> DT32714
R517: step 5 set default values

(SUB 2) ch0HomeSeq

711 R536 ch0 home trig — [F1 DMV , K 5000 , DT 5004] Ch0'w's peed

719 R536 ch0 home trig — R511 <SET> ch0h step1; R536 <RST> ch0 home trig

726 R511 ch0h step1 / X18 ch0 sensor / R509 ch0h step 2 — R504 [ ] Ch0JogL

730 R504 Ch0JogL — [F1 DMV , H 113 , DT 5000] Ch0 con code
[F1 DMV , DT 5004 , DT 5002] Ch0'w's peed, Ch0'r'speed
[F172 PLSH , DT 5000 , K 0] Ch0 con code

750 X18 ch0 sensor / R511 ch0h step1 / R509 ch0h step 2 — R509 <SET> ch0h step 2; R511 <RST> ch0h step1

759 R509 ch0h step 2 — [F1 DMV , K 500 , DT 5004] Ch0'w's peed

767 ch0 home step 2: index right off sensor

```
X18      R509     R514
-| |-----| |-----|/|-------------------------------------------------------> 1
ch0 sen  ch0h ste ch0h ste
sor      p 2      p3

1 ---->-[F1  DMV      ,  H  103     ,  DT  5100   ]
                                     ch0con
                                     code

        [F1  DMV      ,  K  3000    ,  DT  5102   ]
                                     ch0initS
                                     peed

        [F1  DMV      ,  K  5000    ,  DT  5104   ]
                                     ch0max
                                     Speed

        [F1  DMV      ,  K  200     ,  DT  5106   ]
                                     ch0acce
                                     l

        [F1  DMV      ,  K-1270     ,  DT  5108   ]
                                     ch0tarV
                                     al

        [F1  DMV      ,  K  0       ,  DT  5110   ]
                                     ch0puls
                                     e stop

        [F171  SPDH   ,  DT  5100   ,  K  0       ]
                       ch0con
                       code
```

817

```
X18      R509     R514     R911C                                    R509
-|/|----| |-----|/|-----|/|-------------------------------------<RST>
ch0 sen  ch0h ste ch0h ste ch0 sig                                  ch0h ste
sor      p 2      p3                                               p 2
                                                                    R514
-                                                                <SET>
                                                                    ch0h ste
                                                                    p3
```

828 ch0 home step 3, jog slowly left until sensor turns on

```
R514     X18                                                        R513
-| |-----|/|-------------------------------------------------------[ ]
ch0h ste ch0 sen                                                    home le
p3       sor                                                        ft 2ndch
```

```
831   jog command 'up' channel1

      R513
      ─┤ ├──[F1  DMV      ,  H  113     ,  DT  5000   ]
      home le                            Ch0 con
      ft 2ndch                            code

            [F1  DMV      ,  DT  5004   ,  DT  5002   ]
                           Ch0'w's       Ch0'r'sp
                           peed          eed

            [F172  PLSH   ,  DT  5000   ,  K  0       ]
                           Ch0 con
                            code

      R514     X18                                                    R514
851   ─┤ ├────┤ ├──────────────────────────────────────────────────────<RST>
      ch0h ste ch0 sen                                                ch0h ste
      p3       sor                                                    p3
                                                                      R515
                                                                      <SET>
                                                                      ch0 step
                                                                      4

859   ch0 home step 4: move to home offset position,
      position set using hmi in 'SET HOME' screen

      R515     X18
      ─┤ ├────┤ ├──────────────────────────────────────────────────────→ 1
      ch0 step ch0 sen
      4        sor

      1 ──→─[F1  DMV      ,  DT  32714  ,  DT  5118   ]
                           ch0hmi        ch0inde
                           Offset        xDist

868   If no comparison, axis will not home when on home sensor

      R515     X18      ┌D <      DT  32714  ,  K  5      ┐
      ─┤ ├────┤ ├──────┤                                  ├──────────────→ 1
      ch0 step ch0 sen  ch0hmi                             ┘
      4        sor      Offset

      1 ──→─[F1  DMV      ,  K  5       ,  DT  5118   ]
                                        ch0inde
                                        xDist

      R515     ┌TMR      8  ,  K      5  ┐
886   ─┤ ├────┤                          ├──────────────────────────────→ 1
      ch0 step                            ┘
      4

      1 ──→─[F3  DMV/     ,  DT  5118   ,  DT  5118   ]
                           ch0inde       ch0inde
                           xDist         xDist
                                                                      R516
            [F23  D+      ,  DT  5118   ,  K  1       ,  DT  5118   ]──[ ]──
                           ch0inde                     ch0inde        ch0hom
                           xDist                       xDist          eOffset
```

```
     math above: DT5118 -> -DT5118
909 -
     R516     X18
    -| |-----| |-------------------------------------------------------------> 1-
     ch0hom   ch0 sen
     eOffset  sor

    -1 ---->-[F1  DMV      ,   H  103     ,   DT  5100    ]
                                           ch0con
                                           code

    -        [F1  DMV      ,   K  3000    ,   DT  5102    ]
                                           ch0initS
                                           peed

    -        [F1  DMV      ,   K  5000    ,   DT  5104    ]
                                           ch0max
                                           Speed

    -        [F1  DMV      ,   K  200     ,   DT  5106    ]
                                           ch0acce
                                           l

    -        [F1  DMV      ,   DT  5118   ,   DT  5108    ]
                           ch0inde         ch0tarV
                           xDist           al

    -        [F1  DMV      ,   K  0       ,   DT  5110    ]
                                           ch0puls
                                           e stop

    -        [F171  SPDH   ,   DT  5100   ,   K  0        ]
                           ch0con
                           code
     R516     X18      R911C                                                  R515
958 -| |-----|/|------|/|-------------------------------------------------<RST>-
     ch0hom   ch0 sen  ch0 sig                                               ch0 step
     eOffset  sor                                                            4
                                                                             R517
    -                                                                      <SET>-
                                                                             ch0h ste
                                                                             p 5
```

968 ch0 home step 5, set default values for index pos/speed/accel

R517 ch0h step 5 — [F1 DMV , K 10000 , DT 5112] ch0initSpeed

[F1 DMV , K 15000 , DT 5114] ch0maxSpeed

[F1 DMV , K 300 , DT 5116] ch0acc/deac

[F1 DMV , K 8000 , DT 5118] ch0indexDist

[F1 DMV , K 50000 , DT 90348] ch0 count

1004 R517 ch0h step 5 — TMR 9 , K 5 — R517 <RST> ch0h step 5; R400 <RST> ch0homing

1014 (RET)

1015 Above 'ret': end of sub2, ch0 homed
Below sub 3: homing sequence - ch1
R411: step 1 jog left, hit sensor
R409: step 2 index right
R414: step 3 jog slower left, hit sensor end
R415: step 4 move to offset set by hmi -> DT32712
R417: step 5 set default values

(SUB 3) ch1HomeSeq

1016 R436 ch1 home trig — [F1 DMV , K 5000 , DT 4004] Ch1'w's peed

1024 R436 ch1 home trig — R411 <SET> ch1h step1; R436 <RST> ch1 home trig

1031 R411 ch1h step1 — X17 ch1 sensor (/) — R409 ch1h step 2 (/) — R404 [ ] Ch1JogU

```
1035 R404 ─┤ ├─[F1 DMV , H 122 , DT 4000 ]
     Ch1Jog U                    Ch1 con code
     [F1 DMV , DT 4004 , DT 4002 ]
                Ch1'w's peed     Ch1'r'speed
     [F172 PLSH , DT 4000 , K 1 ]
                Ch1 con code

1055 X17 ─┤ ├─ R411 ─┤ ├─ R409 ─┤/├──────────────
```

```
1122  X17      R409     R414     R911D                                          R409
      -|/|-    -| |-    -|/|-    -|/|-                                         <RST>
      ch1 sen  ch1h ste ch1h ste ch1 sig                                        ch1h ste
      sor      p 2      p3                                                      p 2
                                                                                R414
                                                                                <SET>
                                                                                ch1h ste
                                                                                p3
      ch1 home step 3, jog slowly up until sensor turns on
1133

      R414     X17                                                              R413
      -| |-    -|/|-                                                           -[ ]-
      ch1h ste ch1 sen                                                          home U
      p3       sor                                                              p 2ndch
      jog command 'up' channel1
1136

      R413
      -| |-[F1  DMV        ,   H  122       ,   DT  4000     ]
      home U                                   Ch1 con
      p 2ndch                                   code

        [F1  DMV        ,   DT  4004     ,   DT  4002     ]
                            Ch1'w's          Ch1'r'sp
                            peed             eed

        [F172  PLSH     ,   DT  4000     ,   K  1         ]
                            Ch1 con
                             code
1156  R414     X17                                                              R414
      -| |-    -| |-                                                           <RST>
      ch1h ste ch1 sen                                                          ch1h ste
      p3       sor                                                              p3
                                                                                R415
                                                                                <SET>
                                                                                ch1 step
                                                                                4
      ch1 home step 4: move to home offset position,
1164  position set using hmi in 'SET HOME' screen

      R415     X17                                                              1
      -| |-    -| |-                                                           ->
      ch1 step ch1 sen
      4        sor

      1 ->-[F1  DMV        ,   DT  32712    ,   DT  4118     ]
                                               ch1inde
                                               xDist
      If no comparison, axis will not home when on home sensor
1173

      R415     X17      D <     DT  32712    ,   K  5         1
      -| |-    -| |-    -[      ]                            ->
      ch1 step ch1 sen
      4        sor

      1 ->-[F1  DMV        ,   K  5         ,   DT  4118     ]
                                               ch1inde
                                               xDist
```

```
1191  R415                                                                  R416
      ch1 step                                                              ch1hom
      4                                                                     eOffset
1193  R416      X17
      ch1hom    ch1 sen
      eOffset   sor                                                         1
      1 ---> [F1  DMV      ,  H  103     ,  DT  4100   ]
                                          ch1Con
                                          Code
             [F1  DMV      ,  K  3000    ,  DT  4102   ]
                                          ch1initS
                                          peed
             [F1  DMV      ,  K  5000    ,  DT  4104   ]
                                          ch1max
                                          Speed
             [F1  DMV      ,  K  200     ,  DT  4106   ]
                                          ch1acce
                                          l
             [F1  DMV      ,  DT  4118   ,  DT  4108   ]
                            ch1inde       ch1tarV
                            xDist         al
             [F1  DMV      ,  K  0       ,  DT  4110   ]
                                          ch1puls
                                          e stop
             [F171  SPDH   ,  DT  4100   ,  K  1       ]
                            ch1Con
                            Code
1242  R416      X17       R911D                                             R415
      ch1hom    ch1 sen   ch1 sig                                           <RST>
      eOffset   sor                                                         ch1 step
                                                                            4
                                                                            R417
                                                                            <SET>
                                                                            ch1h ste
                                                                            p 5
```

```
1252  ch1 home step 5, set default values for index pos/speed/accel

      R417
      ─┤ ├──[F1  DMV      ,  K 10000   ,  DT  4112   ]
      ch1h ste                            ch1initS
      p 5                                 peed
            [F1  DMV      ,  K 15000   ,  DT  4114   ]
                                          ch1max
                                          Speed
            [F1  DMV      ,  K 300     ,  DT  4116   ]
                                          ch1acc/
                                          deacc
            [F1  DMV      ,  K 8000    ,  DT  4118   ]
                                          ch1inde
                                          xDist
            [F1  DMV      ,  K 50000   ,  DT  90352  ]
                                          ch1 cou
                                          nt
      R417     TMR      0  ,  K      5
1288  ─┤ ├──┬──────────────
```

```
1319 R104 ─┤ ├─[F1  DMV      ,  H  1122    ,  DT  2000   ]
     Ch2Jog                                  Ch2 con
     L                                        code
     ─        [F1  DMV      ,  DT  2004    ,  DT  2002   ]
                              Ch2'w's        Ch2'r'sp
                              peed           eed
     ─        [F172  PLSH   ,  DT  2000    ,  K  2       ]
                              Ch2 con
                               code
     X16      R111     R109                                          R109
1339 ─┤ ├──────┤ ├──────┤/├──────────────────────────────────────────<SET>
     ch2 sen  ch2h ste ch2h ste                                      ch2h ste
     sor      p1       p 2                                          p 2
     ─                                                               R111
                                                                     <RST>
                                                                     ch2h ste
                                                                     p1
     R109
1348 ─┤ ├─[F1  DMV      ,  K  500     ,  DT  2004   ]
     ch2h ste                                Ch2'w's
     p 2                                     peed
     ch2 home step 2: move off sensor, set index parameters
1356 ─

     X16      R109     R114
     ─┤ ├──────┤ ├──────┤/├──────────────────────────────────────────────────────────────────────────────────────→ 1
     ch2 sen  ch2h ste ch2h ste
     sor      p 2      p3
     1 ──→─[F1  DMV      ,  H  1103    ,  DT  3000   ]
                                          ch2Con
                                          Code
     ─        [F1  DMV      ,  K  3000    ,  DT  3002   ]
                                          ch2initS
                                          peed
     ─        [F1  DMV      ,  K  5000    ,  DT  3004   ]
                                          ch2max
                                          Speed
     ─        [F1  DMV      ,  K  200     ,  DT  3006   ]
                                          ch2acce
                                          l
     ─        [F1  DMV      ,  K  1270    ,  DT  3008   ]
                                          ch2tarV
                                          al
     ─        [F1  DMV      ,  K  0       ,  DT  3010   ]
                                          ch2puls
                                          e stop
     ─        [F171  SPDH   ,  DT  3000    ,  K  2       ]
                              ch2Con
                              Code
```

```
1406  X16      R109     R114     R911E                                          R109
      ch2 sen  ch2h ste ch2h ste ch2 sig                                        <RST>
      sor      p 2      p3                                                      ch2h ste
                                                                                p 2
                                                                                R114
                                                                                <SET>
                                                                                ch2h ste
                                                                                p3
      ch2 home step 3, jog slowly left until sensor turns on
1417

      R114     X16                                                              R113
      ch2h ste ch2 sen                                                          home L
      p3       sor                                                              2ndch2
      R113
1420  home L
      2ndch2   [F1  DMV        ,   H  1122      ,   DT  2000     ]
                                                   Ch2 con
                                                    code
               [F1  DMV        ,   DT  2004     ,   DT  2002     ]
                                   Ch2'w's          Ch2'r'sp
                                   peed             eed
               [F172  PLSH     ,   DT  2000     ,   K  2         ]
                                   Ch2 con
                                    code
      R114     X16                                                              R114
      ch2h ste ch2 sen                                                          <RST>
      p3       sor                                                              ch2h ste
                                                                                p3
                                                                                R115
                                                                                <SET>
                                                                                ch2 step
                                                                                4
      ch2 home step 4: move to home offset position,
1448  position set using hmi in 'SET HOME' screen

      R115     X16                                                              1
      ch2 step ch2 sen
      4        sor
      1 ---->  [F1  DMV        ,   DT  32710    ,   DT  3018     ]
                                   hmiCh2           ch2inde
                                   offset           xDist
      If no comparison, axis will not home when on home sensor
1457

      R115     X16      D <        DT  32710    ,   K  5         1
      ch2 step ch2 sen
      4        sor                hmiCh2
                                  offset
      1 ---->  [F1  DMV        ,   K  5         ,   DT  3018     ]
                                                   ch2inde
                                                   xDist
      R115                                                                      R116
1475  ch2 step                                                                  ch2hom
      4                                                                         eOffset
```

```
1477  R116      X16
      ch2homeOffset  ch2 sensor                                                  1
      1 ---> [F1 DMV      ,  H 1103     ,  DT 3000   ]
                                        ch2ConCode
             [F1 DMV      ,  K 3000     ,  DT 3002   ]
                                        ch2initSpeed
             [F1 DMV      ,  K 5000     ,  DT 3004   ]
                                        ch2maxSpeed
             [F1 DMV      ,  K 200      ,  DT 3006   ]
                                        ch2accel
             [F1 DMV      ,  DT 3018    ,  DT 3008   ]
                          ch2indexDist  ch2tarVal
             [F1 DMV      ,  K 0        ,  DT 3010   ]
                                        ch2pulse stop
             [F171 SPDH   ,  DT 3000    ,  K 2       ]
                          ch2ConCode
1526  R116      X16      R911E                                                    R115
      ch2homeOffset  ch2 sensor  ch2 sig                                         <RST>
                                                                                 ch2 step 4
                                                                                 R117
                                                                                 <SET>
                                                                                 ch2h step 5
      ch2 home step 5, set default values for index pos/speed/accel
1536
      R117
      ch2h step 5  [F1 DMV      ,  K 10000    ,  DT 3012   ]
                                        ch2initSpeed
             [F1 DMV      ,  K 15000    ,  DT 3014   ]
                                        ch2maxSpeed
             [F1 DMV      ,  K 300      ,  DT 3016   ]
                                        ch2acc/deacc
             [F1 DMV      ,  K 5080     ,  DT 3018   ]
                                        ch2indexDist
             [F1 DMV      ,  K 50000    ,  DT 90356  ]
                                        ch2 count
```

```
1572  R117   TMR     2 ,  K    5 ]
      ch2h ste                                                  R117
      p 5                                                       <RST>
                                                                ch2h ste
                                                                p 5
                                                                R402
                                                                <RST>
                                                                ch2 ho
                                                                ming

1582                                                            (   RET   )

1583  Above: end of sub 4
      Below: end of sub 1

                                                                (   RET   )

1584  Sub 5:
      If magazine is empty or too full -> set empty, skip calculation
      laser measurement calculation in program: y = 2.73x + 2125
      laser measurement calculation physical:
      ch 1 servo travel distance (mm) = [(0.0273mm) x laser reading] + 21.25mm

      Later add adjustment in the panel for 2125, if picker is going to deep decrease
      this #, if too shallow increase this number

                                                                (SUB    5)

1585  R156   D <     DT  2100  ,  K  780 ]
      calcCh1                                                   R157
      dist      laser val                                       <SET>
                ue                                              empty
      R156   D >     DT  2100  ,  K  3850 ]
      calcCh1                                                   R156
      dist      laser val                                       <RST>
                ue                                              calcCh1
                                                                dist
                                                                R103
                                                                <RST>
                                                                notEmp
                                                                ty
```

```
1615  R156      D >      DT 2100   , K 779      D <      DT 2100   , K 3851
      calcCh1   laser val                        laser val
      dist      ue                               ue                          1

      1 ---> [F31 D*     , DT 2100   , K 273     , DT 2102   ]
                         laser val               ch1 calc
                         ue

             [F33 D%     , DT 2102   , K 100     , DT 2102   ]
                         ch1 calc                ch1 calc

             [F23 D+     , DT 2102   , DT 32722  , DT 2102   ]
                         ch1 calc    laserOff    ch1 calc
                                     set

             [F1 DMV     , DT 2102   , DT 4118   ]
                         ch1 calc    ch1inde
                                     xDist

             [F3 DMV/    , DT 2102   , DT 2102   ]
                         ch1 calc    ch1 calc

             [F23 D+     , DT 2102   , K 1       , DT 4120   ]
                         ch1 calc                -ch1ind
                                                 exDista

                                                                R157
                                                                <RST>
                                                                empty
                                                                R156
                                                                <RST>
                                                                calcCh1
                                                                dist
                                                                R103
                                                                <SET>
                                                                notEmp
                                                                ty

1701                                                            ( RET )

1702  Sub 6: Ch 1 index down based off of laser measurement in Sub 5

                                                                (SUB 6)
```

```
1703 R90
     ─┤ ├──[F1  DMV      ,  H  103     ,  DT  4100   ]
     indexDo                            ch1Con
     wn                                 Code
     ├       [F1  DMV      ,  K  60000   ,  DT  4102   ]
                                        ch1initS
                                        peed
     ├       [F1  DMV      ,  K  78000   ,  DT  4104   ]
                                        ch1max
                                        Speed
     ├       [F1  DMV      ,  K  200     ,  DT  4106   ]
                                        ch1acce
                                        l
     ├       [F1  DMV      ,  DT  4118   ,  DT  4108   ]
                        ch1inde         ch1tarV
                        xDist           al
     ├       [F1  DMV      ,  K  0       ,  DT  4110   ]
                                        ch1puls
                                        e stop
                                                                    R90
     ├       [F171  SPDH   ,  DT  4100   ,  K  1       ]──────────<RST>──
                        ch1Con                                      indexDo
                        Code                                        wn
1754 ──────────────────────────────────────────────────────────(    RET    )──

     Sub 7: Ch 1 index up based off of laser measurement in Sub 5
1755 ├
     ──────────────────────────────────────────────────────────(SUB     7)──
```

```
1756 R91
     ─┤ ├──[F1  DMV      ,  H  103     ,  DT  4100   ]
     indexUp                             ch1Con
                                         Code
          [F1  DMV      ,  K  60000   ,  DT  4102   ]
                                         ch1initS
                                         peed
          [F1  DMV      ,  K  78000   ,  DT  4104   ]
                                         ch1max
                                         Speed
          [F1  DMV      ,  K  200     ,  DT  4106   ]
                                         ch1acce
                                         l
          [F1  DMV      ,  DT  4120   ,  DT  4108   ]
                          -ch1ind        ch1tarV
                          exDista        al
          [F1  DMV      ,  K  0       ,  DT  4110   ]
                                         ch1puls
                                         e stop
                                                                          R91
          [F171  SPDH   ,  DT  4100   ,  K  1       ]──────────────────<RST>
                          ch1Con                                          indexUp
                          Code
1807 ──────────────────────────────────────────────────────────────(    RET    )──

     Sub 8: Ch 0 index right 2"
1808
     ──────────────────────────────────────────────────────────────(SUB     8)──
```

```
1809 R92
     ─┤ ├──[F1  DMV      ,  H  103     ,  DT  5100   ]
     indexRi                            ch0con
     ght                                code
          [F1  DMV      ,  K  20000   ,  DT  5102   ]
                                        ch0initS
                                        peed
          [F1  DMV      ,  K  95000   ,  DT  5104   ]
                                        ch0max
                                        Speed
          [F1  DMV      ,  K  450     ,  DT  5106   ]
                                        ch0acce
                                        l
          [F1  DMV      ,  K-5080     ,  DT  5108   ]
                                        ch0tarV
                                        al
          [F1  DMV      ,  K  0       ,  DT  5110   ]
                                        ch0puls
                                        e stop
                                                                        R92
          [F171  SPDH   ,  DT  5100   ,  K  0       ]──────────────────<RST>
                         ch0con                                         indexRi
                         code                                           ght
1860 ──────────────────────────────────────────────────────────────────(    RET    )──

     Sub 9: Ch 0 index left 2"
1861 ─
     ──────────────────────────────────────────────────────────────────(SUB      9)──
```

```
1862  ch0 index left
      R93
      ─┤ ├─[F1  DMV      ,  H  103     ,  DT  5100   ]
      indexLeft                          ch0concode
      [F1  DMV      ,  K  10000   ,  DT  5102   ]
                                         ch0initSpeed
      [F1  DMV      ,  K  95000   ,  DT  5104   ]
                                         ch0maxSpeed
      [F1  DMV      ,  K  450     ,  DT  5106   ]
                                         ch0accel
      [F1  DMV      ,  K  5080    ,  DT  5108   ]
                                         ch0tarVal
      [F1  DMV      ,  K  0       ,  DT  5110   ]
                                         ch0pulse stop
                                                                    R93
      [F171  SPDH   ,  DT  5100   ,  K  0       ]──────────────────<RST>
                       ch0concode                                   indexLeft
1913  ──────────────────────────────────────────────────────────────(    RET    )──
      Sub 10: Ch 2 index right 2"
1914
      ──────────────────────────────────────────────────────────────(SUB    10)──
```

```
1915 R95
     ch2indexR
     ─┤ ├──[F1  DMV      ,  H  1103    ,  DT  3000   ]
                                         ch2ConCode
     [F1  DMV      ,  K  10000   ,  DT  3002   ]
                                         ch2initSpeed
     [F1  DMV      ,  K  40000   ,  DT  3004   ]
                                         ch2maxSpeed
     [F1  DMV      ,  K  200     ,  DT  3006   ]
                                         ch2accel
     [F1  DMV      ,  K  5080    ,  DT  3008   ]
                                         ch2tarVal
     [F1  DMV      ,  K  0       ,  DT  3010   ]
                                         ch2pulse stop
     [F171  SPDH   ,  DT  3000   ,  K  2       ]──────────────────────────────R95
                      ch2ConCode                                              <RST>
                                                                              ch2indexR
1966 ──────────────────────────────────────────────────────────────────────( RET )─

     Sub 11: Ch 2 index left 2"
1967

     ──────────────────────────────────────────────────────────────────────(SUB  11)─
```

```
      R96
1968 ─┤ ├──[F1  DMV      ,  H  1103    ,  DT  3000   ]
      ch2inde                            ch2Con
      xL                                 Code
           [F1  DMV      ,  K  10000   ,  DT  3002   ]
                                         ch2initS
                                         peed
           [F1  DMV      ,  K  40000   ,  DT  3004   ]
                                         ch2max
                                         Speed
           [F1  DMV      ,  K  200     ,  DT  3006   ]
                                         ch2acce
                                         l
           [F1  DMV      ,  K-5080     ,  DT  3008   ]
                                         ch2tarV
                                         al
           [F1  DMV      ,  K  0       ,  DT  3010   ]
                                         ch2puls
                                         e stop
                                                                          R96
           [F171  SPDH   ,  DT  3000   ,  K  2       ]──────────────────<RST>──
                            ch2Con                                        ch2inde
                            Code                                          xL
2019 ──────────────────────────────────────────────────────────────(   RET   )──
```